

5.7.3

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Question: For the matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 2 & -1 & 3 \end{pmatrix},$$

show that

$$A^3 - 6A^2 + 5A + 11I = 0. \quad (0.1)$$

Find A^{-1} .

Solution: The characteristic equation is

$$\left| \begin{pmatrix} 1-\lambda & 1 & 1 \\ 1 & 2-\lambda & -2 \\ 2 & -1 & 3-\lambda \end{pmatrix} \right| = 0, \quad (0.2)$$

which expands to

$$\lambda^3 - 6\lambda^2 + 5\lambda + 11 = 0. \quad (0.3)$$

By Cayley-Hamilton theorem,

$$A^3 - 6A^2 + 5A + 11I = 0. \quad (0.4)$$

Rearranging,

$$11I = -A^3 + 6A^2 - 5A, \quad (0.5)$$

and multiplying both sides by A^{-1} ,

$$A^{-1} = \frac{1}{11}(-A^2 + 6A - 5I) \quad (0.6)$$

$$= \frac{1}{11} \begin{pmatrix} -3 & 4 & 4 \\ 7 & 0 & -3 \\ 5 & -3 & 0 \end{pmatrix}. \quad (0.7)$$

Final Answer:

$$A^{-1} = \begin{pmatrix} -\frac{3}{11} & \frac{4}{11} & \frac{4}{11} \\ \frac{7}{11} & 0 & -\frac{3}{11} \\ \frac{5}{11} & -\frac{3}{11} & 0 \end{pmatrix}. \quad (0.8)$$